Muhammad Abusaqer

Fulbright Alumnus

Assistant Professor (tenure-track), Computer Science & Cybersecurity, Minot State University

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Artifacts: Project pages & selected materials (datasets, tools, slides)

Research Lab: Trustworthy Language Intelligence Lab (TLI Lab)

Research Summary

- **Goal:** build an externally funded research program in trustworthy NLP for security-relevant and harmful online text, emphasizing incident-centric discourse triage and prioritization.
- **Systems and artifacts:** CyberTweetGrader&Labeler (CTGL) for cyber-incident discourse prioritization; auditable dataset curation and quality measurement.
- **Methods:** reproducible benchmarking across traditional ML, transformers, and prompted LLMs; scalable, auditable pipelines for incident intelligence.

Keywords: security-aware NLP; trustworthy LLM evaluation; incident-centric OSINT triage (CTGL); human-in-the-loop labeling & dataset curation; online abuse/cyberbullying detection.

Current Academic Appointment

Minot State University, Minot, ND — Assistant Professor (tenure-track) of Computer Science & Cybersecurity (Aug 2022–present)

- **Teaching**: Computer Networks, Defensive Network Security, Vulnerability Analysis, Ethical Hacking, Mobile and Wireless Security, Applied Cryptography, Web & Internet Programming, CS II, Software Engineering & Testing; Operating Systems (Spring 2026). Modalities: in-person & online; platforms: Blackboard Ultra; zyBooks/JBL Cloud Labs; Pearson MyLab, Revel; Cengage MindTAP, SAM.
- **Research**: AI/ML and NLP for security-relevant text; domain-specific evaluation; curated datasets; student mentoring (16 undergraduate coauthors, 2022–2026); externally funded internal grants and proposal activity.
- **Service**: Faculty Senate; Academic Assessment Committee; programming competition coaching (DigiKey); department faculty hiring committee.

Education

- Ph.D., Computer Science (Cybersecurity & AI), North Dakota State University, Fargo, ND, 2025. Dissertation: CyberTweetGrader&Labeler: A Domain-Specific NLP Approach for Mining and Classifying Cybersecurity Discourse on Social Media. Advisor: Prof. Simone Ludwig.
- M.S., Computer Science and Applications, Virginia Tech, Blacksburg, VA, 2005.
- Diploma in Education (two-year, post-baccalaureate teacher-education program; completed concurrently with B.S. coursework), Birzeit University, 1995.
- B.S., Mathematics, Birzeit University, 1995. *Class rank: 2nd*; GPA 3.6/4.0.

Awards and Honors

- **FULBRIGHT FELLOWSHIP** — competitive national selection (Palestine), awarded 2001; funded M.S. study at Virginia Tech (2003–2005).
- NDSU Teaching Assistantship (2015–2021): Covered tuition and a monthly stipend.
- National Merit Scholarship (1990–1995): Full tuition support and a monthly stipend.
- Undergraduate class rank: 2nd in B.S. Mathematics, Birzeit University (1995).

Grants and Funding (Selected)

- 2026 · NSF CISE Future Computing Research (Future CoRe) — Collaborative proposal in trustworthy NLP for security-relevant text · PI; collaborative proposal with S. Ludwig · Pending.
- 2025–2026 · Minot State Univ. Faculty Small Grant — Expanding and Validating CyberTweetGrader&Labeler · PI & proposal author · $4,000 · Funded.
- 2023 · Minot State Univ. Empower Grant — Cybersecurity curriculum enhancement · PI & proposal author · $6,000 · Funded.
- 2023–2024 · ND EPSCoR — Cybersecurity NLP for the regional industry · PI & proposal author · $40,000 · Not funded.
- 2015/2016 · Bank of Palestine (Zamalah) — Faculty development grant · Recipient & proposal author · $30,000 · Awarded.
- 2014 · Qatari Aid Association — New campus building grant (infrastructure) · Proposal coauthor · $1,200,000 · Awarded.
- 2008 · International Bank (via Palestinian Ministry of Education) — Strategic plan development grant · Proposal lead/coauthor · $100,000 · Awarded.
- 2008 · AMIDEAST & U.S. Dept. of Education — Fulbright Alumni Development Grant · Recipient & proposal author · $3,000 · Awarded.

Publications

Refereed conference papers († student mentee)

[1] **M. Abusaqer** and J. Saquer, "A Comparative Analysis of Transformer and Traditional ML Models for Cyberbullying Detection on Twitter (now X)," in Proc. 2025 *IEEE 49th Annual Computers, Software, and Applications Conference* (*COMPSAC*), Toronto, ON, Canada, 2025, pp. 1607–1612. doi: 10.1109/COMPSAC65507.2025.00216.

[2] **M. Abusaqer**, M. B. Senouci, and K. Magel, "Twitter User Sentiments Analysis: Health System Cyberattacks Case Study," in Proc. 2023 *International Conference on Artificial Intelligence in Information and Communication* (*ICAIIC*), Bali, Indonesia, 2023, pp. 018–024. doi: 10.1109/ICAIIC57133.2023.10067026.

[3] A. Pun†, B. Olson†, and **M. Abusaqer**, "Predicting Student Academic Performance: Using Machine Learning and Clustering," in Proc. 57th *Midwest Instruction and Computing Symposium (MICS)*, Minneapolis, MN, USA, Apr. 4–5, 2025.

[4] D. Degele† and **M. Abusaqer**, "Analyzing Ransomware Incidents in Healthcare: Patterns and Risk Assessment," in Proc. 57th Midwest Instruction and Computing Symposium (MICS), Minneapolis, MN, USA, Apr. 4–5, 2025.

[5] *T. Khan*†, and **M. Abusaqer**, "Evaluating Quick-Commerce Platforms: A Sentiment and Topic Modeling Analysis of User Reviews," in Proc. *57th Midwest Instruction and Computing Symposium (MICS)*, Minneapolis, MN, USA, Apr. 4–5, 2025.

[6] *J. Jensen*† and **M. Abusaqer**, "Global Echoes of the FIFA World Cup 2022: Sentiment and Theme Analysis via Deep Learning and Machine Learning on Twitter," in Proc. *56th Midwest Instruction and Computing Symposium (MICS)*, Minneapolis, MN, USA, Apr. 5–6, 2024.

[7] *S. Khan*†, *K. Khan*†, and **M. Abusaqer**, "Text Detection between an AI-Written Passage vs. a Human-Written Passage," in Proc. *56th Midwest Instruction and Computing Symposium (MICS)*, Minneapolis, MN, USA, Apr. 5–6, 2024.

[8] *T. Smith*† and **M. Abusaqer**, "Predicting Campus Crime Based on State Firearm Policy," in Proc. *56th Midwest Instruction and Computing Symposium (MICS)*, Minneapolis, MN, USA, Apr. 5–6, 2024.

[9] **M. Abusaqer** and *C. Fofie*†, "Cyberbullying Classification Using Three Deep Learning Models: GPT, BERT, and RoBERTa," in Proc. *55th Midwest Instruction and Computing Symposium (MICS)*, Cedar Falls, IA, USA, Mar. 31–Apr. 1, 2023.

[10] **M. Abusaqer** and *Q. Sullivan*†, "Darknet Traffic Classification Using Deep Learning," in Proc. *55th Midwest Instruction and Computing Symposium (MICS)*, Cedar Falls, IA, USA, Mar. 31–Apr. 1, 2023.

[11] *A. Scott*†, *J. T. Snow*†, and **M. Abusaqer**, "Automated Categorization of Cybersecurity News Articles through State-of-the-Art Text Transfer Deep Learning Models," in Proc. *55th Midwest Instruction and Computing Symposium (MICS)*, Cedar Falls, IA, USA, Mar. 31–Apr. 1, 2023.

[12] **M. Abusaqer** and K. Magel, "Comparison of Students' Learning and Engagement on an Online Course Before and After the Spread of COVID-19," in Proc. *54th Midwest Instruction and Computing Symposium (MICS)*, Milwaukee, WI, USA, Apr. 1–2, 2022, pp. 251–257.

[13] **M. Abusaqer** and K. Magel, "Teaching Computer Packages to Students Different in Everything," in Proc. 52nd *Midwest Instruction and Computing Symposium (MICS)*, Fargo, ND, USA, Apr. 5–6, 2019.

Technical report

[14] Y. Bhardwaj, **M. Abu-Saqer**, and M. A. Pérez-Quiñones, "General Interface Description of Websites Using CLICK and UIML." Department of Computer Science, Virginia Tech. Tech. Rep. TR-04-27, 2004.

Accepted / forthcoming conference papers († student mentee)

[A1] T. Khan†, D. Alonso†, and **M. Abusaqer**, "Empirical Evaluation of Data Poisoning Attacks and Practical Defenses in Supervised Learning," to appear in Proc. 58th Midwest Instruction and Computing Symposium (MICS), Eau Claire, WI, USA, Mar. 27–28, 2026; accepted for in-person presentation at MICS 2026; full paper to appear in the conference proceedings.

[A2] W. Novak† and **M. Abusaqer**, "Empirical Evaluation of Membership Inference Attacks on NLP Text Classifiers: A Baseline Study on SST-2," to appear in Proc. 58th Midwest Instruction and Computing Symposium (MICS), Eau Claire, WI, USA, Mar. 27–28, 2026; accepted for in-person presentation at MICS 2026; full paper to appear in the conference proceedings.

[A3] P. Hummel†, R. Skabo†, and **M. Abusaqer**, "Evasion Attacks: How Adversarial Noise Bypasses ML Classifiers," to appear in Proc. 58th Midwest Instruction and Computing Symposium (MICS), Eau Claire, WI, USA, Mar. 27–28, 2026; accepted for in-person presentation at MICS 2026; full paper to appear in the conference proceedings.

Manuscripts Under Review / Under Revision / In Preparation

- [15] **M. Abusaqer**, "A controlled evaluation of prompted LLM inference vs. fine-tuned encoders for text classification under reliability and cost constraints," submitted to ICML 2026, Jan. 2026 (under review).
- [16] **M. Abusaqer**, "Green NLP for Online Abuse Detection: Accuracy–Latency–Energy Trade-offs of TF-IDF (Logistic Regression, Linear SVM) vs. Fine-Tuned Transformers," manuscript under revision for resubmission.
- [17] **M. Abusaqer**, "Caption-then-Classify for Multimodal Harmful Meme Detection: A Lightweight Baseline for Social-Good," manuscript under revision for resubmission.
- [18] **M. Abusaqer** and S. Ludwig, "A Curated Twitter/X Dataset of the 2020 UHS Ransomware Incident: Collection Protocols, Ethics, and Incident-Response Utility," submitted to IEEE Access, Dec. 2025 (under review).
- [19] **M. Abusaqer**, "TREST: Trust-Calibrated Relevance Scoring for Healthcare Cyber-Incident Tweets," manuscript in preparation.
- [20] **M. Abusaqer** and S. Ludwig, "CyberTweetGrader&Labeler: A Domain-Specific NLP System for Prioritizing Cyber-Incident Discourse on Twitter/X," manuscript in preparation for submission to IEEE Open Journal of the Computer Society.

Research Projects

Selected Research Projects and Contributions (PI-led and recent)

- CyberTweetGrader&Labeler (CTGL): domain-specific NLP pipeline for prioritizing cyber incident discourse on Twitter/X; reproducible security-aware feature engineering, mathematically grounded relevance scoring, and labeling with an accompanying artifact package. Status: in submission/in preparation [18]–[20]. Related prior work: [2].

- Cyberbullying and abusive-language detection on Twitter/X: comparative evaluation of transformers, traditional ML, and prompted LLM inference; analysis of reliability, cost, and deployment trade-offs. Published: COMPSAC 2025 [1]. Submitted: ICML 2026 [15].
- Energy-aware benchmarking for online abuse detection: joint evaluation of accuracy, throughput/latency, and GPU energy per inference for TF-IDF baselines vs. fine-tuned transformers. Current status: manuscript under revision for resubmission [16].
- Multimodal harmful meme detection: training-free caption-then-classify baseline for Hateful Memes with calibration-aware decision rules and efficiency analysis. Current status: manuscript under revision for resubmission [17].
- Health-system cyberattacks sentiment and theme analysis: event-focused analysis around healthcare cyber incidents with time-aware collection and ML baselines. Published: ICAIIC 2023 [2].
- COVID-era learning engagement: pre-/post-COVID comparison of engagement signals in an online course using machine learning. Published: MICS 2022 [12].
- Web UI abstractions (CLICK/UIML): technical report on interface description and transformation. VT CS TR-04-27 [14].
- Transportation AI at UGPTI: applied computer vision and AI to North Dakota roadway analytics, including vehicle detection/counting, data preparation, model development/testing, and a prototype distinguishing 13 vehicle classes in support of transportation research and decision support.
- Research-computing enablement at CCAST/NDSU: authored user-facing tutorials and C-shell automation for the Thunder and Thunder Prime HPC systems; supported researchers in biology, chemistry/biochemistry, computer science, and related fields in using high-performance computing for ongoing research workflows.

Selected Student-Mentored Projects

- Membership inference attacks on NLP text classifiers: baseline study on SST-2. MICS 2026 accepted / forthcoming in proceedings.
- Data poisoning attacks and practical defenses in supervised learning. MICS 2026 accepted / forthcoming in proceedings.
- Evasion attacks: adversarial-noise robustness demonstrations for machine-learning classifiers. MICS 2026 accepted / forthcoming in proceedings.
- Predicting student academic performance: supervised learning and clustering for academic risk and grouping. MICS 2025 [3].
- Ransomware in healthcare: pattern mining and risk stratification using public incident reports. MICS 2025 [4].
- Global Echoes of the FIFA World Cup 2022: sentiment and theme analysis on Twitter with DL/ML. MICS 2024 [6].
- Text detection (AI vs. human): empirical evaluation of LLM-generated vs. human-written text using lexical and syntactic baselines. MICS 2024 [7].
- Automated categorization of cybersecurity news: transfer-learning pipeline to triage security-news articles for threat-intel workflows. MICS 2023 [11].

Selected Presentations

- M. **Abusaqer**, "CyberTweetGrader&Labeler: Social Media Analytics for Cyberattack Intelligence," Oral presentation, ND EPSCoR Annual State Conference, NDSU Memorial Union, Fargo, ND, Oct. 21, 2025.
- M. **Abusaqer**, "EVPN-VXLAN Made Teachable: Datacenter Networking Labs for Early Undergraduates," Poster presentation, ND EPSCoR Annual State Conference, NDSU Memorial Union, Fargo, ND, Oct. 21, 2025.

Teaching (Courses And New Preps)

Minot State University (2022–present)

- **Courses**: CSCI 330 Software Engineering & Testing; CSCI 340 Computer Networks; CSCI 410 Defensive Network Security; CSCI 415 Vulnerability Analysis; CSCI 420 Mobile & Wireless Security; CSCI 425 Applied Cryptography; CSCI 221 Web & Internet Programming; CSCI 161 CS II; Operating Systems (Spring 2026).
- **Modalities and breadth**: in-person, hybrid, and online at scale; upper-division teaching areas include ethical hacking/network analysis, applied cryptography, secure web/mobile systems, defensive network security, vulnerability analysis, and threat-intelligence/security-analytics topics.
- **Services**: Programming Competition Coach (2022–present): Weekly training and coaching for collegiate programming contests, including DigiKey Collegiate Computing Competition (DKC^3) and the MICS Programming Contest.

Previous Academic Appointments

- North Dakota State University (NDSU), Fargo, ND — Adjunct Professor, Management Information Systems (MIS), Accounting, Finance, and MIS Dept. (Aug 2017–Aug 2021).
    - Led "Business Use of Computers," large cohorts (450 fall & 350 spring), applied labs; supervised graduate TAs for labs, grading, and logistics.
- Teaching Assistant, Computer Science Department (2015–2021).
- Al-Aqsa University (public university in Palestine) — Tenure-track Junior Lecturer/Instructor, Computer Science Dept. (Dec 2006–Aug 2015).
    - Taught: Artificial Intelligence; Computer Science I & II; Operating Systems; Web Programming.
    - Led personal initiative to the deployment of the university's first Moodle-based LMS (installation/configuration, operational support, faculty training, documentation).
- University College of Applied Sciences (UCAS), Palestine — Instructor, Computers & Networking Dept. (2000–2006).
    - Taught: Linux; Windows Server administration; Intro to Programming; Operating Systems; authored Arabic-language materials; organized computing forums.

Earlier Technical and Industry Experience

- Systems Programmer & Analyst — ATS (Arab Technology Systems; acquired by Paltel), Ramallah, Palestine (Jul 1995–Sep 2000). Paradox/FoxPro/Oracle development, maintenance, and performance optimization.
- Network Administrator & Technical Support (student employment) — Birzeit University IT Division, Palestine (Mar 1993–Jul 1995). Novell NetWare administration; lab LAN support; early email-system installation.

Research Mentoring († Student Mentee)

16 undergraduate coauthors, 2022–2026

- A. Pun†; B. Olson† — Student success prediction via ML and clustering (MICS 2025).
- D. Degele† — Ransomware in healthcare: patterns and risk (MICS 2025).
- T. Khan† — Quick-commerce platforms: sentiment & topic modeling (MICS 2025); data poisoning attacks and practical defenses in supervised learning (MICS 2026 accepted / forthcoming, with D. Alonso†).
- W. Novak† — Membership inference attacks on NLP text classifiers: baseline study on SST-2 (MICS 2026 accepted / forthcoming).
- P. Hummel†; R. Skabo† — Evasion attacks and adversarial robustness demonstrations across image and text classifiers (MICS 2026 accepted / forthcoming).
- J. Jensen† — FIFA World Cup 2022: sentiment & themes (MICS 2024).
- S. Khan†; K. Khan† — AI vs human text detection (MICS 2024).
- T. Smith† — Campus crime vs firearm policy (MICS 2024).

- C. Fofie† — Cyberbullying with GPT/BERT/RoBERTa (MICS 2023).
- Q. Sullivan† — Darknet traffic classification (MICS 2023).
- A. Scott†; J. T. Snow† — Cybersecurity news categorization (MICS 2023).

Professional Service

University and Departmental

- Faculty Senate (2024–present).
- Academic Assessment Committee (2024–present).
- Department Hiring Search Committee (2023–present).
- Leadership roles at Al-Aqsa University: Head of CS department; Vice Dean of Planning & Development; Assistant to the Dean of Graduate Studies for Research & Scholarships.

Scholarly

- Reviewer, ICML (2026–present).
- Reviewer, IEEE Access (2025–present).
- Reviewer, IEEE Open Journal of the Computer Society (2025–present).
- Reviewer, "Applied AI for Public Safety and Security" workshop at IEEE BigData 2025.
- Reviewer, IEEE BigData 2025 special session/workshop: Big Data & ML in Healthcare (Dec 8–11, 2025).
- Session Chair: ML for Business, MICS 2025; AI Classification, MICS 2024; Deep Learning, MICS 2023.
- Co-coordinator, Minot State AI & Data Summit (Apr. 29, 2024).

Community

- ChatGPT and AI Systems Presentation, Lions Club (2023).
- Fulbright Alumni speaker & organizer (2006–2014).
- American Graduates Association in Palestine & Palestinian Fulbright Alumni Association: leadership board and events (2006–2014).

Research Artifacts And Planned Releases

- CyberTweetGrader&Labeler (planned release): reproducible pipeline and evaluation scripts for cyber-incident tweet prioritization; dataset schema and sample pack.
- UHS 2020 ransomware-incident dataset (planned release): curated Twitter/X posts with documentation, schema, and reproducibility artifacts.

Professional Development And Certifications

- SEED Workshop, Hands-On Labs for Security Education, Syracuse University (2018).
- CIW Web Security Associate (2013).
- Microsoft Certified: Azure Fundamentals (2021).
- National Cyber League (NCL) Competitor (2021).
- Certification in Teaching Undergraduates, NDSU (2016).
- Data Scientist's Toolbox, Johns Hopkins Univ. via Coursera (2019).
- Cisco CCNA (2006).
- Microsoft Certified System Engineer (MCSE), Windows Server (2002).
- Oracle Database OCP Certification: 8i Certified Database Administrator (2002).
- LinkedIn Training Courses: ongoing completions in AI, ML, and cybersecurity.

Professional Memberships

- IEEE
- ACM

Technical Skills

- Programming: Python, Java, C++.
- ML/DL: PyTorch, TensorFlow; scikit-learn; experiment tracking & evaluation.
- NLP: spaCy, NLTK, Gensim, Hugging Face (BERT, RoBERTa, GPT-2/3.5).
- Security/Networking: Wireshark, Nmap, Kali/Metasploit; network security/IDS concepts.
- Data & Web: SQL, basic web stack (HTML/CSS/JS), RESTful APIs.
- LMS/EdTech: Blackboard, Moodle, YuJa/Verity.

References

Available upon request.